

tromagnetic pulse which is oscillating at a specific frequency. The more frequent the oscillation the higher the frequency and vice versa. The data is transmitted on these oscillating radio waves of different frequencies and require a device capable of reading them at the specific frequency. Otherwise these waves simply pass through objects without any data transmission.

The measurement of the frequencies is done on the basis of cycles per second e.g. 1MHz is a frequency of 1 million cycles per second, where MHz is megahertz. The current Internet infrastructure is built on two separate RF frequencies which are 2.4Ghz to 2.48GHz and 5.15GHz to 5.85GHz. These are typically referred to as 2.4GHz and 5GHz bands respectively. The former band is free to use by anyone, anywhere, for any purpose which makes it free but inevitably clogged if used across vast distances due to Internetwork conflicts. The current Wi-Fi infrastructure protocol of 802.11, cordless phones, newer walkie-talkies, wireless intercoms, automatic doors, microwave ovens, emergency radios and even Bluetooth is on the 2.4GHz band. In contrast, the 5GHz band is far more wider and relatively uncluttered by different device types. It is used in some consumer products but is largely the space where the Internet of Things is intended to grow.

The Wi-Fi protocol is powering modern wireless networks already and has been standardised across the globe by the IEEE (Institute of Electrical and Electronics Engineers) and the Wi-Fi Alliance. However, the standard is split across numerous 802.11 protocols which offer a range of different performances. The most current of which is the 802.11ac which primarily uses the 5GHz band but also works on the 2.4GHz band. This allows the range of a 802.11ac device to easily work across the expanse of a whole home unit. The Wi-Fi enabled router or hub is the common node for all devices and streams data through it to other devices. This means that when a user uses a tablet on the home network to push a video to the TV, it passes through the router or hub first and then goes to the TV. More the devices , more the work intensive the router becomes. This initial setup of all the devices requires a one-time manual step but with hundreds or thousands of “things” it can become untenable in the future. This places Wi-Fi in a dubious category when considering it as a primary source of networking in the age of the Internet of Things.

Within shorter distances the inter-device connectivity can be done using the very popular Bluetooth connection. Already used for smartphone accessories and media systems, it works efficiently within smaller locations. It also operates in the 2.4GHz range but doesn't require a central hub or